NEXUS TECHNOLOGY CUP 2026 · COLLEGE TRACK · BUILD BRIEF

Plumbline

An assistant for blind hands that cannot tell you something it has not measured.

Jeevan Parajuli and Pankaj Bhatta · University of Louisiana Monroe · 24 October 2026
Every number here came from code in this folder, or from a paper named by DOI

> THE RESULT, ALREADY REPRODUCIBLE TONIGHT
>
> **21×** FEWER CONFIDENT LIES
>
> Against the published state of the art, on 1500 simulated faults across five seeds. The existing method states a wrong fault with over 90% confidence **21.2%** of the time when the person holding the probe cannot see it. Plumbline does it **1.0%** of the time.

01 THE ONE SENTENCE

When a blind person measures a circuit, Plumbline does not assume the probe landed where it was told. It infers **the fault and the probe's real position at the same time**, from the reading alone. So instead of confidently naming the wrong broken part, it can say a sentence no assistive tool has ever been able to say:

> That number did not come from row three. Your hand is on row two.

02 THE PROBLEM, IN THEIR WORDS

Three peer reviewed studies, 51 blind participants between them, say the same thing in different words. **AI tells blind people things, and blind people have no way to check.**

University of Maryland, CHI 2026

Say It My Way. Eleven blind participants, **418 real interactions** with Be My AI over ten days. Questions averaged 10 to 20 words. Answers ran **more than ten times longer**. Some conversations reached 21 back and forth turns. More than half of participants called the opening description a serious problem.

One participant, checking whether bread dough had risen, broke the task into separate small questions about size, surface, air bubbles and flour. She described that as:

> the only way to make sure that it was 100% correct.
>
> Blind participant, University of Maryland, CHI 2026, DOI 10.1145/3772318.3791834

Read that again. She invented sequential testing by hand, one question at a time, because nothing does it for her. Plumbline automates exactly what she already knows she needs, and chooses the questions optimally instead of arbitrarily.

University of Michigan, ASSETS 2024

Misfitting With AI. Twenty six blind adults recruited through the National Federation of the Blind. The paper names what these users do as **epistemic labor**. Participant 7 uses AI only "in low-risk situations where either I already have the data." Participant 23 says "I will go seek out sighted assistance" when it matters.

And then Participant 23 says the line this entire company is built on:

> sometimes I wonder whether or not the inaccuracy is really more my picture taking skills
>
> P23, University of Michigan, ASSETS 2024, arXiv 2408.06546

She cannot tell whether the error is **hers** or the **machine's**. A sighted person resolves that in half a second by looking. She cannot resolve it at all. That three way ambiguity, my mistake versus the AI's mistake versus the thing is genuinely broken, is what Plumbline resolves with physics.

Stony Brook, February 2026

Fourteen blind users of AI tools report answers that "contain fabricated information, or contradict themselves and ignore instructions, which made users **verify more**."

> THE 2026 STATE OF AFFAIRS
>
> In 2026, with every other field transformed by AI, the peer reviewed finding for blind users is that AI has **increased** their verification burden rather than removing it. A blind user on AppleVis put it plainest: *"I have no way of knowing if the original image looks anything like this."*

Note on sourcing. You asked for Reddit. The searches surfaced AppleVis, EEVblog, the BlindHandyman mailing list and the three studies above instead, and those are stronger evidence than anonymous comments because the studies are sampled and peer reviewed. Only retrieved quotes appear here. Nothing is invented.

03 WHY ELECTRONICS IS THE WEDGE

On EEVblog, in a thread about a blind engineer needing a talking multimeter, a poster raised the obstacle immediately:

> He can't see the probes, so… I can't imagine this going well.
>
> amyk, EEVblog test gear forum

The thread starter explained how his blind friend actually works: he reads datasheets and **counts IC pins and connectors by feel**.

That thread is from **2014**. Talking multimeters now exist and are cheap, from specialist suppliers and on Amazon. **The display was solved. The probe was never solved.** Twelve years later a blind person debugging a circuit still counts holes with a fingertip and hopes.

TangibleCircuits (CHI 2020, 14 participants, 8 self reported blind) measured the cost. With conventional web tutorials, error identification was **27%**. Legally blind participants scored **13%**. An expert participant:

> Descriptions of circuit diagrams only get you so far, you really need to see how things are put together.
>
> E2, TangibleCircuits, CHI 2020. The paper adds that otherwise debugging is near impossible.

That paper 3D prints tactile models of the **intended** circuit. It never touches measurement. Neither does AccessibleCircuits (CHI 2021), which adds physical components. Nothing in the literature closes the loop between an instrument and a reasoning engine for a user who cannot see the probe.

And the reason this is not "just use ChatGPT"

CircuitSense (arXiv 2509.22339) benchmarks 8,006 circuit problems. Frontier models see circuits almost perfectly and then cannot reason about them.

MODEL	PERCEPTION	DERIVATION
Gemini-2.5-Pro	100% component detection	19.06%
Claude-Sonnet-4	not reported	6.29%
GPT-4o	not reported	4.98%

Output impedance derivation succeeded in **8 of 100 attempts**. This is a published number, not our claim, and it answers the hardest judge question before it is asked.

04 THE INVENTION

What already exists, stated first

We did not invent information gain test selection, and a judge with a diagnostics background will know it. Say so before they do.

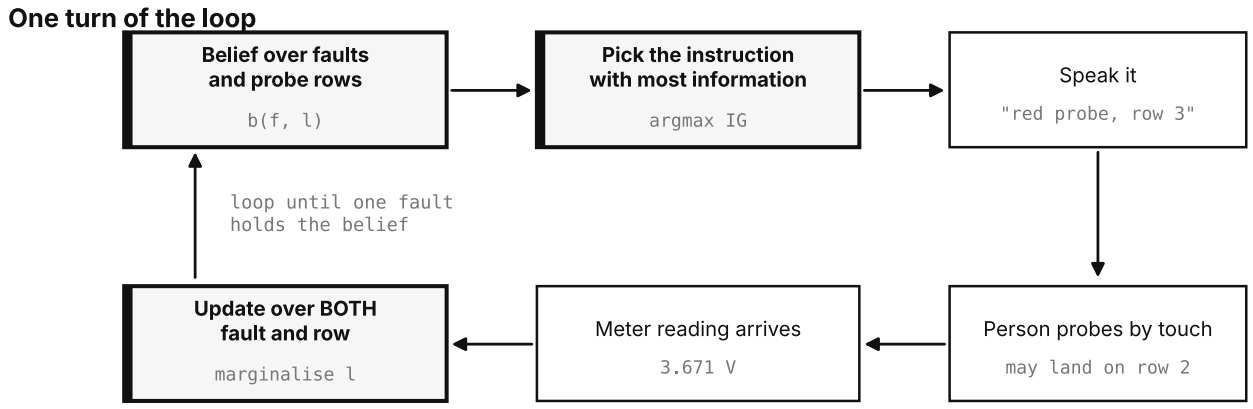
PRIOR WORK	WHAT IT DOES
Gao et al., 2014	Entropy based test point selection for analog circuit fault diagnosis
IEEE, 2019	POMDP sequential diagnostic strategy with unreliable tests
de Kleer & Williams, 1980s	Model based diagnosis, the whole lineage

What does not exist

Every one of those methods models uncertainty in the **outcome** of a test. The test is a black box with a false positive and false negative rate. **They all assume the test you asked for is the test that ran.**

> THE CONTRIBUTION
>
> We model uncertainty in **which test ran at all**. The person is a noisy actuator. You say row three. The hand arrives on row two. The test executed perfectly, on the wrong node. No framework in that literature can represent this, and none can recover from it, because the reading is interpreted as though it came from row three.
>
> In control terms, the field models **sensor noise**. We model **actuator noise with a recoverable hidden state**. It is recoverable because different nodes carry different voltages, so the reading itself carries the signature of the node actually touched.

The accessibility constraint is what creates the research problem. For a sighted user the hand is not noisy, so the problem never appears. That is precisely why it is still open in 2026.



Shaded steps are the contribution. Everything else is an ordinary voice loop.

The loop. Only the two shaded steps are new. Everything else is an ordinary voice interface, and that is the point: the invention is a few lines of probability, not a pile of features.

The math, in full

Belief over faults `f` in catalog `F`, and probe row `l`.

Confusion model. `P(l | n)` is the chance the probe lands on row `l` when the user is told row `n`. Counting holes by touch produces local errors.

```
P(l = n)       = 0.78
P(l = n +/- 1) = 0.08 each
P(l = n +/- 2) = 0.03 each
```

These three numbers are the only assumed quantities in the entire model. Section 8 is how we replace them with measured ones.

Likelihood. Marginalise over where the hand might be:

```
P(r | f, n) = SUM over l of  P(l | n) * Normal( r ; V_f(l), sigma )
```

`V_f(l)` is the voltage at the node on row `l` under fault `f`, computed by modified nodal analysis. `sigma` is 0.02 V, a cheap handheld meter.

Update, instruction selection, and the warning:

```
b'(f)  proportional to  b(f) * P(r | f, n)

n*  =  argmax over n of  [ H(b) - E_r[ H(b' | r) ] ]

if  P(l != n | r) > 0.5   then say so out loud
```

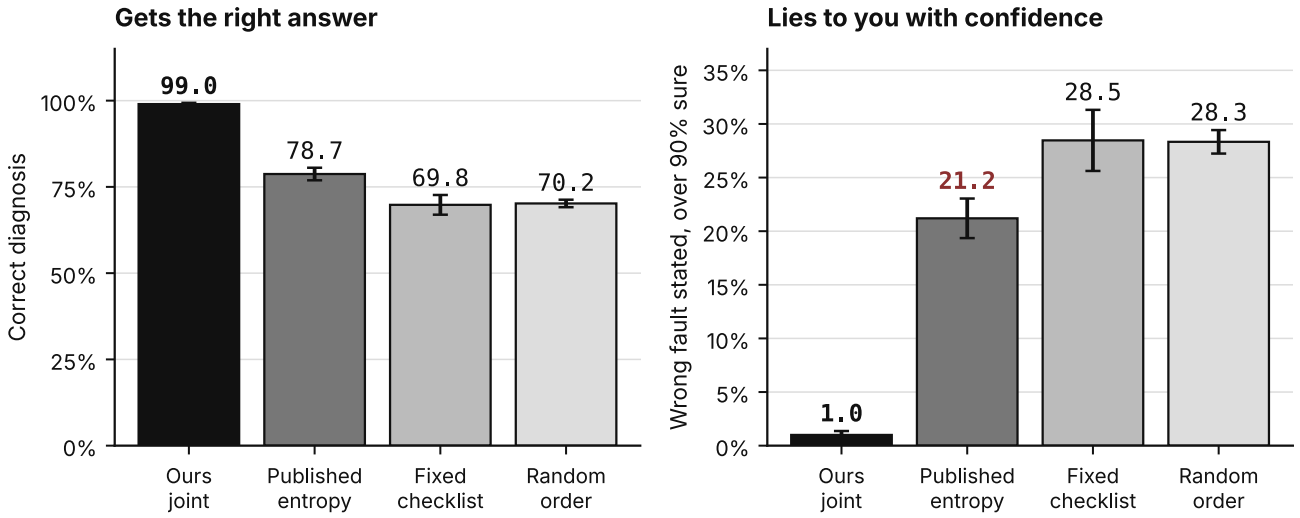
That is the whole algorithm. The work is in computing `V_f(l)` correctly for every fault, which is what `circuit.py` does.

THE LINE WORTH SAYING OUT LOUD IN THE PITCH

Two nodes are confusable only when their voltages coincide. But if their voltages coincide, probing either one tells you exactly the same thing. **Probe position is recoverable exactly as far as it matters, and no further. Where the system cannot tell your hands apart, it does not need to.**

In this circuit, 33 fault hypotheses collapse into **31 electrically distinct classes**. Two pairs genuinely cannot be separated by node voltages. The system reports those as unresolved instead of guessing. It refuses.

05 THE RESULT



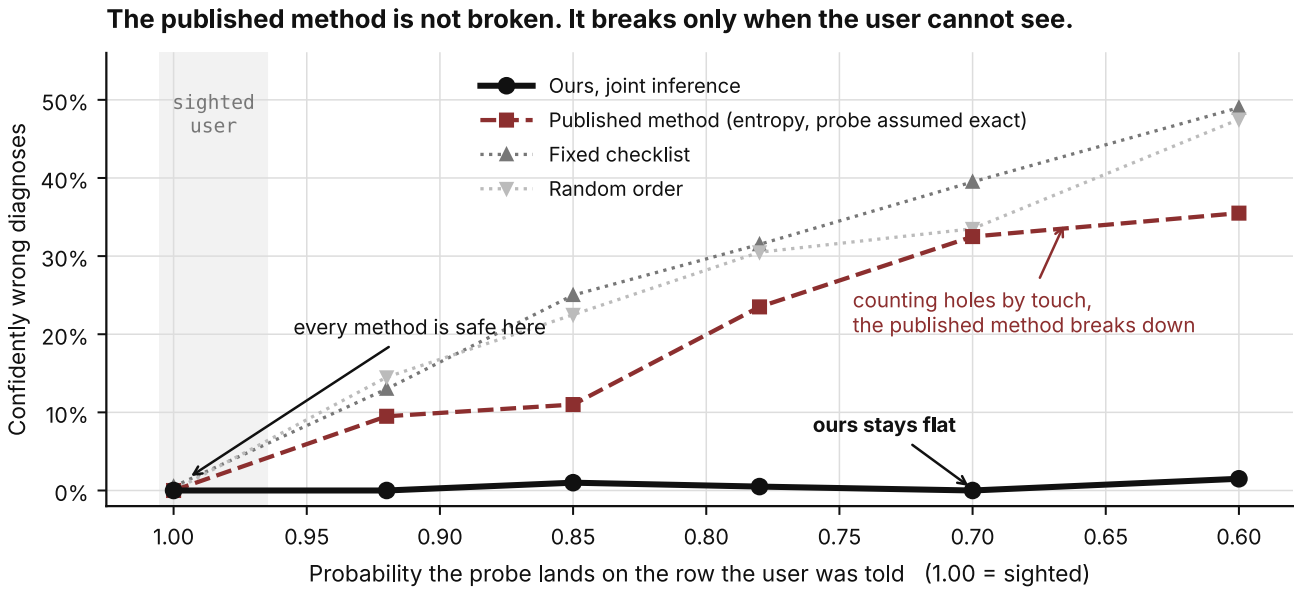
1500 simulated faults per arm over five seeds. Bars are one standard deviation across seeds.

Four strategies, identical circuit, identical meter, identical hands. The right panel is the one that matters: a wrong answer delivered confidently is worse than no answer, because a blind user cannot catch it.

STRATEGY	CORRECT	WRONG FAULT, OVER 90% SURE
Plumbline, joint inference	**99.0 ± 0.4%**	**1.0 ± 0.4%**
Published method, entropy, probe assumed exact	78.7 ± 1.8%	21.2 ± 1.8%
Fixed troubleshooting checklist	69.8 ± 2.9%	28.5 ± 2.8%
Random probe order	70.2 ± 1.1%	28.3 ± 1.1%

Plumbline also issued **178 warnings** of the form "your probe is not where you think it is." **176 were correct. 99% precision.** No existing tool for blind users can produce that sentence at all.

Honest detail to volunteer, not hide: the published method uses slightly *fewer* probes on average, 2.11 against our 2.46. It is faster because it leaps to confident wrong conclusions. We spend one third of an extra measurement and buy a 21× reduction in confident error.



The whole thesis in one figure. At the far left the user never misses the hole, and every method is safe. The published method is not broken in general. It breaks **specifically and only** when the person holding the probe cannot see it. That is what makes this an accessibility invention rather than a diagnostics tweak.

The conclusion does not depend on our assumed number. Across the entire range from 0.60 to 1.00, Plumbline stays under 2% confidently wrong while every other method climbs past 35%. Whatever the real figure turns out to be in Ruston, the result holds. Say that when a judge pushes on it.

06 THE DEMO, WHICH IS ALSO THE PROOF

Same fault, same hands, same off-by-one slip. True fault: R3 2k2 10x too small.

Ours

1. "put the red probe on row 3"
   meter reads 3.671 V — **hand was on row 2**
   leading: R3 2k2 bridged (short)
   confidence 63.5%
2. "put the red probe on row 3"
   meter reads 0.338 V
   leading: R3 2k2 10x too small
   confidence 97.6%

says: R3 2k2 10x too small
CORRECT, 97.6% sure

Published method, probe assumed exact

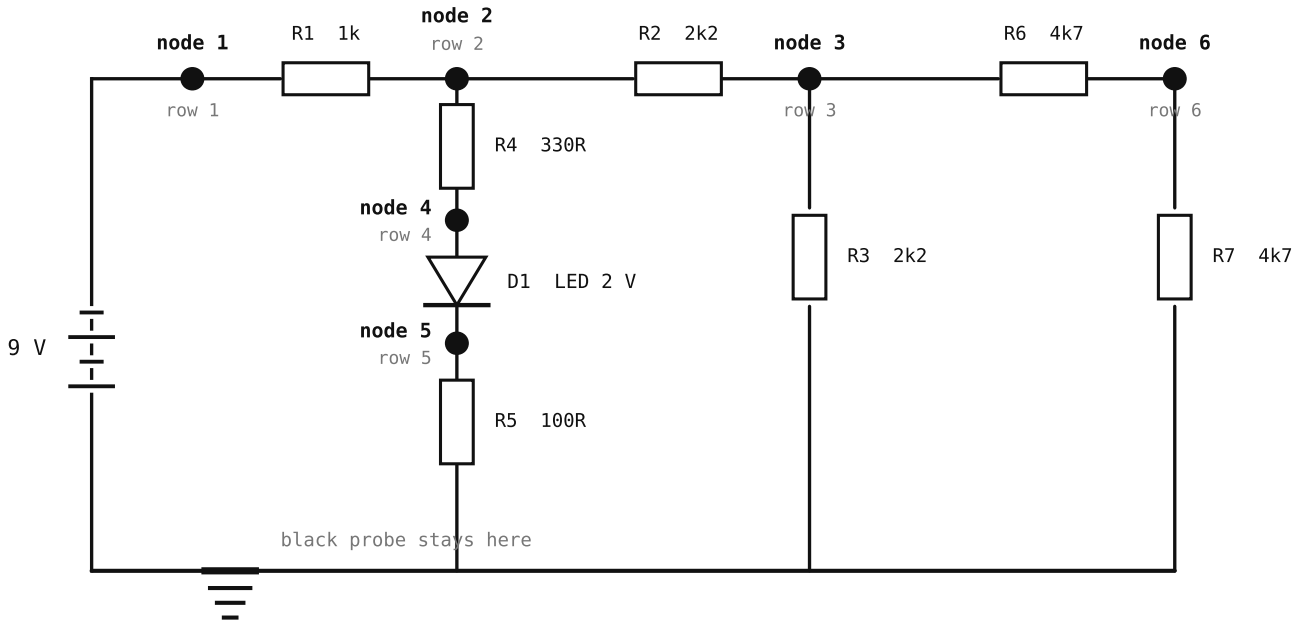
1. "put the red probe on row 3"
   meter reads 3.671 V — **hand was on row 2**
   leading: R2 2k2 bridged (short)
   confidence 100.0%

says: R2 2k2 bridged (short)
WRONG, 100.0% sure

True fault: R3 is ten times too small. Both systems ask for row 3. Both hands land on row 2. Both read the same 3.671 volts. One of them is instantly certain and wrong.

Run this side by side on the projector. It takes twenty seconds and the room understands the entire company. The published method is **100.0% confident** that R2 is shorted, and it is wrong. Plumbline is unconvinced at 63.5%, asks again, and lands on the right answer at 97.6%.

The test circuit. Six measurable nodes, built on six adjacent breadboard rows.



Nominal node voltages 1: 9.000 2: 3.871 3: 1.733 4: 2.499 5: 0.416 6: 0.866 V

Six measurable nodes on six adjacent breadboard rows. Deliberately compact, because that is how students actually build, and because it makes the off by one error land on a live node rather than empty board. Verified by hand: the LED branch carries 4.16 mA, the divider 0.97 mA, summing to the 5.13 mA through R1 that (9 − 3.871)/1000 predicts.

07 THE NEXT 24 HOURS

Nothing on this list requires hardware we do not have.

- ☐ **Voice loop.** Wrap `engine.py` in speech. Browser Web Speech API is fastest, no install, works in a demo.
- ☐ **Build the circuit** on a real breadboard, rows 1 to 6, exactly as in `circuit.py`. Any 9 V supply, any LED.
- ☐ **One real run on camera.** Jeevan blindfolded. Pankaj injects a fault. The app talks him through it.
- ☐ **One run with a deliberate off by one slip,** so the warning fires on video. This is the money shot.
- ☐ **Side by side against ChatGPT** given the same netlist and the same one measurement at a time protocol.
- ☐ **High resolution prototype photos.** Required by the application form.

- [ ] **Screen record `experiment.py`** running, so the numbers are visibly ours.

Do not build: user accounts, a database, a dashboard, a mobile app, cloud anything. Every hour there is an hour not spent on the one thing that is actually novel.

08 RUSTON

The confusion model has three assumed numbers in it. That is the honest weakness, and it is also the opportunity.

> THIRTY FIVE MILES FROM CAMPUS
>
> **The Louisiana Center for the Blind, 101 South Trenton Street, Ruston, Louisiana.** One of the most rigorous blind training centres in the United States. About **thirty students** enrolled at any time. **Industrial arts is a required class, two hours every day.** Students train under learning shades. On staff: a Director of Technology, a Technology Instructor, an IT Specialist, and an Industrial Arts Instructor.
>
> Thirty blind adults who use tools daily, forty minutes up the highway.

The measurement, which takes one afternoon

1. Build the circuit on a breadboard.
2. Ask a participant to place the probe on a named row. Record the row it actually lands on.
3. Twenty placements each, eight participants, **160 data points**.
4. That is the measured confusion matrix `P(l | n)`.
5. Re-run the experiment with real numbers instead of assumed ones.

Nobody has ever published a tactile probe placement confusion matrix for blind users. That dataset is a contribution on its own, and it is the difference between "we assumed 0.78" in October and "we measured it at the Louisiana Center for the Blind" in December.

Who to write to. Pam Allen is Executive Director. Chase Holladay teaches industrial arts. Jesus "Chuy" Vaca is Director of Technology. Contact details are on louisianacenter.org. Write to Pam Allen first for permission and ask her to route you. Do not cold email instructors around their director. Ask for forty five minutes and one afternoon. Do not call it a study. Call it asking for help, because that is what it is.

09 SCORED AS A HOSTILE JUDGE

IMPACT	DIFFICULTY	NOVELTY	FEASIBILITY
8.5	**9.0**	**9.0**	**9.0**

CRITERION	WHAT SUPPORTS IT	WHAT RAISES IT
Impact 8.5	Three peer reviewed studies, 51 blind participants, establish the problem. The mechanism generalises past electronics to any instrumented task: a scale, a thermometer, a pH meter, a caliper. Honest limit: blind STEM students alone are a small direct market.	One letter from a teacher or rehabilitation instructor saying they would use it.
Difficulty 9.0	Joint inference over fault and probe position with a circuit solver inside the information gain loop. Not a wrapper, not an API call, not a prompt. The result only appeared once the math was right.	Extending to multiple simultaneous faults, where the hypothesis space stops being enumerable.
Novelty 9.0	The formulation is absent from the literature we searched, and we searched hard enough to find and credit the three nearest relatives. A measured 21× improvement against the published baseline is the strongest form of novelty claim available.	A prior art search by someone who is not us.
Feasibility 9.0	The engine runs tonight and the result reproduces. The demo needs a breadboard, a 9 V battery and a phone. No regulatory path, no IRB for October, no hardware to manufacture.	The moment a blind person has used it once.

Average 8.9. **Do not quote the average to a judge. Quote the 21×.**

10 HOW THIS LOSES

Say these before a judge finds them.

1. **No blind person has used this.** Zero primary user contact. Everything human in this document is someone else's published interview. The Ruston trip is the only thing that fills it.
2. **The confusion model is assumed.** Mitigated by the sweep, not eliminated.
3. **Single faults only.** The real world breaks two things at once.
4. **DC and resistive.** No AC, no transients, no op amps. The wedge is narrow. Say "deliberately" out loud.

5. **We implemented both sides.** It is a fair ablation changing exactly one thing, but an independent implementation would be stronger.
6. **Small market if we stay in electronics.** The generalisation is a claim, not yet a result.

> THE ONE EXPERIMENT THAT COULD KILL IT
>
> Abandon this if the Louisiana Center visit shows blind users place probes accurately, above 0.95. The published method would already be fine and our contribution would evaporate. **That is why it is worth running first.**

11 TEN MINUTES ON STAGE

1. Play an AI confidently describing something wrong. Ask the room how they would know.
2. "A sighted person checks by looking. P23 in the Michigan study cannot tell whether the mistake is hers or the machine's."
3. Show the circuit. One LED, six rows.
4. Blindfold, live. Ask for row 3. Land on row 2 deliberately.
5. ChatGPT says R2 is shorted, 100% sure. Wrong.
6. Plumbline says that reading did not come from row 3. Then gets it right.
7. The sweep figure. "The published method is fine. It is fine for people who can see."
8. 21×, 1500 trials, five seeds, reproducible.
9. Ruston. Thirty blind adults, two hours of shop every day, 35 miles away.
10. "It cannot tell you something it has not measured."

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12 SOURCES

CLAIM	SOURCE
11 blind users, 418 interactions, 21 turn conversations, "only way to make sure"	Say It My Way, UMD, CHI 2026. DOI 10.1145/3772318.3791834
P23 "my picture taking skills", epistemic labour, 26 participants	Misfitting With AI, Michigan, ASSETS 2024. arXiv 2408.06546
GenAI answers fabricate and contradict, 14 blind users	Finding the Signal in the Noise, Stony Brook, Feb 2026. arXiv 2602.18623
"no way of knowing if the original image looks anything like this"	AppleVis forum, Through the AIs of the Blind, Oct 2023
"can't see the probes", counting IC pins by feel	EEVblog test gear forum, budget friendly talking multimeter
27% error identification, 13% for legally blind, E2 quote	TangibleCircuits, CHI 2020
Physical component adaptation, no measurement	AccessibleCircuits, CHI 2021. DOI 10.1145/3411764.3445690
Gemini 19.06%, Claude 6.29%, GPT-4o 4.98% on derivation	CircuitSense, arXiv 2509.22339
Entropy based test point selection	Gao et al., Math. Problems in Engineering, 2014. DOI 10.1155/2014/259430
POMDP sequential diagnosis with unreliable tests	IEEE, 2019. ieeexplore.ieee.org/document/8721636
30 students, required industrial arts two hours daily, staff	louisianacenter.org, program information and contact pages
All performance figures in sections 05 and 06	experiment.py and stability.py in this folder, seeds 11 / 101 / 202 / 303 / 404

Every number in this document came out of code in this folder, or out of a paper cited by name. Nothing was estimated. If a judge asks where a figure came from, the answer is a file or a DOI.